

```

<program> ::= <statement> | <statement> <program>
<statement> ::= <init> | <input> | <add> | <assign> | <map> | <filter> | <sort> | <unique> | <join>
| <output> | <if> | <for>
<init> ::= "list" <id> "=" <list_literal>
<input> ::= "input" <id>
<add> ::= "add" <id> <value>
<assign> ::= <id> "=" <expression>
<map> ::= "map" <id> <var> "=>" <expr>
<filter> ::= "filter" <id> <var> "=>" <expr>
<sort> ::= "sort" <id> <order>
<unique> ::= "unique" <id>
<join> ::= "join" <id> <string>
<output> ::= "output" <id>
<if> ::= "if" <condition> <statements> "end"
<for> ::= "for" <var> "in" <id> <statements> "end"
<statements> ::= <statement> | <statement> <statements>
<expression> ::= <id> <operator> <id> | <id> <operator> <list_literal> | <id> <operator>
<number>
<condition> ::= <expr>
<expr> ::= <var> | <var> <comp_op> <value> | <function> "(" <var> ")" | <expr> <logical_op>
<expr>
<list_literal> ::= "[" <value_list> "]"
<value_list> ::= <value> | <value> "," <value_list>
<value> ::= <string> | <number>
<id> ::= <identifier>
<var> ::= <identifier>
<identifier> ::= <letter> | <letter> <ident_rest>
<ident_rest> ::= <letter> | <digit> | "'" | <letter> <ident_rest> | <digit> <ident_rest> | ""
<ident_rest>
<operator> ::= "+" | "-" | "*"
<comp_op> ::= "==" | "!=" | "<" | "<=" | ">" | ">="
<logical_op> ::= "and" | "or"
<function> ::= "size" | "length"
<order> ::= "asc" | "desc"
<string> ::= "" <string_content> ""
<string_content> ::= <character> | <character> <string_content>
<character> ::= <digit> | <letter>
<number> ::= "0" | <nonzero_digit> <digits_opt>
<digits_opt> ::= <digit> | <digit> <digits_opt> | ""
<nonzero_digit> ::= "1" | "2" | "3" | "4" | "5" | "6" | "7" | "8" | "9"
<digit> ::= "0" | "1" | "2" | "3" | "4" | "5" | "6" | "7" | "8" | "9"
<letter> ::= "a" | "b" | ... | "z" | "A" | "B" | ... | "Z"

```